

CSE370: Database Systems
Spring 2026
Total Marks: 10 Marks
Duration - 20 minutes

Name:

Section:

Student ID:

Q1. There are two types of data independence in database management systems:

Logical Data Independence and Physical Data Independence.

You work for a music streaming company. Due to rapid growth in user traffic, the database administrators have decided to move all user data from HDD storage to SSD storage and store the existing tables across multiple servers to improve performance and availability.

Read the scenario below, then answer the following 3 questions.

- a) What type of **Data Independence** is this? [1 Mark]

Answer: Physical Data Independence

- b) Which Schema does this change in relation to **Three Schema Architecture**? [1 Mark]

Answer: Internal Schema

- c) Name the next step of the affected schema you mentioned in (b) and mention whether it will show any changes due to the given scenario. [1 Mark]

Answer: Conceptual Schema and it will show no changes due to the given scenario

Q2. A Space Research Organization wants to create a database system for managing its missions, scientists, and facilities. The requirements are given below. [7 Marks]

- Scientists are part of a Research Division. Scientists have **unique id**, specialization, name, address, date_joined, and a list of research_skills.
- A Scientist's address is composed of country, state, and city.
- Each Research Division has a **unique name**, email, research_domain, and director.
- Each Scientist must belong to a Research Division, and each Research Division must have many scientists.
- Each Research Division is managed by one Scientist. A Scientist cannot manage more than one division. When a Scientist starts managing a Division, the **starting_date** is **recorded**.
- Scientists can collaborate with other Scientists. Each Scientist can collaborate with more than one colleague. A project name is recorded for each collaboration.
- Scientists also work on Space Missions. Each Mission has a **unique mission_id**, destination, budget, and launch_date.

- Some Research Divisions have Robots for assisting research. Each Robot has a name, model, function, and operating_hours. None of these attributes are guaranteed to be unique, different divisions may even have robots with the same name. However, note that the same Division will not reuse the same name for its own robots.

Now, draw an **ER diagram** of this system.

Note: Write down all assumptions (if any). It is **mandatory to show the cardinality ratio and the participation type for each relationship**.

